

Practice Questions (Chapter 14 - Chapter 19)

CHAPTER 14

Question 1 The Equity Premium and the Value of Stocks. Recall that the current stock price is computed by

$$Q_t = \frac{\$D_{t+1}^e}{(1 + i_{1t} + x)} + \frac{\$Q_{t+1}^e}{1 + i_{1t} + x}$$

- 1) If the risk premium is larger, all else being equal, what happens to the price of the stock today?
- 2) If the one-period interest rate increases, what happens to the price of the stock today?
- 3) If the expected value of the stock at the beginning of period $t + 1$ increases, what happens to the value of the stock today?

Note that the stock pricing formula given above can be expanded as follows up to n years from now.

$$Q_t = \frac{\$D_{t+1}^e}{(1 + i_{1t} + x)} + \frac{\$D_{t+2}^e}{(1 + i_{1t} + x)(1 + i_{1t+1}^e + x)} + \dots$$

$$+ \frac{\$D_{t+n}^e}{(1 + i_{1t} + x) \cdots (1 + i_{1t+n-1}^e + x)} + \frac{\$Q_{t+n}^e}{(1 + i_{1t} + x) \cdots (1 + i_{1t+n-1}^e + x)}$$

- 4) Set $i_{1t} = i_{1t+n} = 0.05$ for all n . Set $x = 0.03$. Compute the coefficients on $\$D_{t+3}^e$ and $\$D_{t+10}^e$. Compare the effect of a \$1 expected increase in a dividend 3 years from now and 10 years from now.
- 5) Repeat the computation in 4) with $i_{1t} = i_{1t+n} = 0.08$ for all n and $x = 0.05$.

CHAPTER 15

Problem 1 Modern Theory of Consumption. A consumer has nonhuman wealth equal to \$100,000. She earns \$40,000 this year and expects her salary to increase by 5% in real terms each year for the following two years. She will then retire. The real interest rate is equal to 0% and is expected to remain at 0% in the future. Labor income is taxed at a rate of 25%

- 1) What is this consumer's human wealth?
- 2) What is the total wealth?
- 3) If she expects to live for seven more years after retiring and wants her consumption **to remain the same** (in real terms) every year from now on, how much can she consume this year?
- 4) If she received a bonus of \$20,000 in the current year only, with all future salary payments remaining as stated earlier, by how much could this consumer increase consumption now and in the future?
- 5) Suppose now that at retirement, Social Security will start paying benefits each year equal to 60% of this consumer's earning during her last working year. Assume that benefits are not taxed. How much can she consume this year and still maintain constant consumption over her lifetime?

CHAPTER 17

Problem 1 Trade Deficit. Consider a world with three equal-sized economies (A,B, and C) and three goods (clothes, cars, and computers). Assume that consumers in all three economies want to spend an equal amount on all three goods. The value of production of each good in the three economies is given in the table below:

	A	B	C
Clothes	10	0	5
Cars	5	10	0
Computers	0	5	10

- 1) What is GDP in each economy? If the total value of GDP is consumed and no country borrows from abroad, how much will consumers in each economy spend on each of the goods?
- 2) If no country borrows from abroad, what will be the trade balance in each country? What will be the pattern on trade in this world (i.e., which good will each country export and to whom)?
- 3) Given your answer to 2), will country A have a zero trade balance with country B? with country C? Will any country have a zero trade balance with any other country?
- 4) The United States has a large trade deficit. If has a trade deficit with each of its major trading partners, but the deficit is much larger with some countries (e.g., China) than with others. Suppose the United States eliminates its overall trade deficit (with the world as a whole). Do you expect it to have a zero trade balance with every one of its trading partners? Does the especially large trade deficit with China necessarily indicate that China does not allow US goods to compete on an equal basis with Chinese goods?

CHAPTER 18

Problem 1 Policy Coordination and the World Economy. Consider an open economy which is characterized by the following behavioral equations:

$$C = 10 + 0.8(Y - T)$$

$$I = 10$$

$$G = 10$$

$$T = 10$$

$$X = 0.3Y^*$$

$$IM = 0.3Y$$

where Y^* denotes foreign output. Assume that the real exchange rate is fixed and equal to one.

- 1) Solve for equilibrium output in the domestic economy, given Y^* . What is the multiplier in this economy? If we were to close the economy—so that exports and imports were identically equal to zero—what would the multiplier be? Why the multiplier be different in a closed economy? Explain.
- 2) Assume that the foreign economy is characterized by the same equations as the domestic economy (with asterisks reversed). Use the two sets of equations to solve for the equilibrium output of each country. (*Hint: Use the equations for the foreign country to solve for Y^* as a function of Y and substitute this solution for Y^* in 1).*) What is the multiplier for each country now? Why is it different from the open economy multiplier in 1)?
- 3) Assume that the domestic government has a target level of output of 125. Assuming that the foreign government does not change G^* , what is the increase in G necessary to achieve the target output in the domestic economy? Solve for net exports and the budget deficit in each country.

- 4) Suppose each government has a target level of output of 125 and that each government increases government spending by the same amount. What is the common increase in G and G^* necessary to achieve the target output in both countries? Solve for net exports and the budget deficit in each country.
- 5) Why is fiscal coordination, such as the common increase in G and G^* in 4), difficult to achieve in practice?

CHAPTER 19

Problem 1 Flexible Exchange Rates and Foreign Macroeconomic Events. Consider an open economy with flexible exchange rates. Let UIP stand for the uncovered interest parity condition.

- 1) In an $IS - LM - UIP$ diagram, show the effect of an increase in foreign output, Y^* , on domestic output (Y) and the exchange rate (E), when the domestic central bank leaves the policy interest rate unchanged.
- 2) In an $IS - LM - UIP$ diagram, show the effect of an increase in the foreign interest rate, i^* , on domestic output (Y) and the exchange rate (E), when the domestic central bank leaves the policy interest rate unchanged.

Problem 2 Flexible Exchange Rates and the Responses to Changes in Foreign Macroeconomic Policy. Suppose there is an expansionary fiscal policy in the foreign country that increases Y^* and i^* at the same time.

- 1) In an $IS - LM - UIP$ diagram, show the effect of the increase in foreign output, Y^* , and the increase in the foreign interest rate, i^* , on domestic output (Y) and the exchange rate (E), when the domestic central bank leaves the policy interest rate unchanged.
- 2) In an $IS - LM - UIP$ diagram, show the effect of the increase in foreign output, Y^* , and the increase in the foreign interest rate, i^* , on domestic output (Y) and the exchange rate (E), when the domestic central bank matches the increase in the foreign interest rate with an equal increase in the domestic interest rate.
- 3) In an $IS - LM - UIP$ diagram, show the required domestic monetary policy following the increase in foreign output, Y^* , and the increase in the foreign interest rate, i^* , if the goal of domestic monetary policy is to leave domestic output (Y) unchanged. When might such a policy be necessary?